

Batch #53 - 6 Apr 2026

SMaSH HBC 431 (V1) - 6.8%

American Pale Ale

Brewer: The Thirsty Otter

Type: All Grain

IBU : 43 (Tinseth)
 BU/GU : 0.86
 Colour : 12 EBC
 Carbonation : 2.5 CO2-vol

Pre-Boil Gravity : 1.037
 Original Gravity : 1.048
 Total Gravity : 1.050
 Final Gravity : 1.009

Fermentables (1.2 kg)

943 g - Chateau Pale Ale 8.5 EBC (78.6%)
 ^ Lot # 5425000394730 (07.01.2025)
 ^ Brouwmaatje (NL) BM-BL.051.613.25/1
 257 g - Chateau Pale Ale 8.5 EBC (21.4%)
 ^ Lot # 5425000394730 (16.01.2026)
 ^ Brouwmaatje (NL) BM-BL.051.613.25/1
 32 g - Bottling - Sugar, Table (Sucrose) 2 EBC
 ^ Albert Heijn (NL)

Hops (46.3 g)

30 min - 4.6 g - HBC 431 - 10% (18 IBU)
 ^ Hopeye (GB) HE-50G-HOPS-HBC431 (50 grams)
 10 min - 7.7 g - HBC 431 - 10% (16 IBU)
 ^ Hopeye (GB) HE-50G-HOPS-HBC431 (50 grams)

Hop Stand

20 min hopstand @ 80 °C
 20 min - 15 g - HBC 431 - 10% (9 IBU)
 ^ Hopeye (GB) HE-50G-HOPS-HBC431 (50 grams)

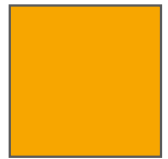
Dry Hops

6 days - 19 g - HBC 431 - 11.4%
 ^ Hopeye (GB) HE-50G-HOPS-HBC431 (50 grams)

01 Brouwpunt 5L (60min) (rev 4)

Batch Size : 5.6 L
 Boil Size : 7.76 L
 Post-Boil Vol : 5.96 L

Mash Water : 3.6 L
 Sparge Water : 5.43 L
 Boil Time : 60 min
 Total Water : 9.03 L



12 EBC

Brewhouse Efficiency: 71.8%
 Mash Efficiency: 73.3%

Mash Profile

01 One Step Mash (60 min)
 71 °C - Strike Temp
 65 °C - 60 min - Temperature

Fermentation Profile

Ale
 20 °C - 14 days - Primary
 20 °C - 45 days - Conditioning

Water Profile

02 NL Spa Reine Flat Mineral Water (www.ah.nl)
 Ca 109 Mg 9 Na 26 Cl 75 SO 233 HCO 17

SO/Cl ratio: 3.1
 Mash pH: 5.26
 Sparge pH: 6

Measurements

Mash pH: 5.27

Boil Volume: 8.25

Pre-Boil Gravity: 1.035

Post-Boil Kettle Volume: 6.35

Original Gravity: 1.052

Fermenter Top-Up:

Fermenter Volume: 4

Final Gravity: 1.005

Bottling Volume: 2.33

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Miscellaneous

Mash - 1.48 g - Calcium Chloride (CaCl₂) 33 %
33%
^ Lot # 115038
^ Brouwstore (NL) 055.035.0
Mash - 0.53 g - Canning Salt (NaCl)
^ Albert Heijn (NL)
Mash - 0.66 g - Epsom Salt (MgSO₄)
^ Lot # /2119000091
^ Brouwstore (NL) 055.027.7
Mash - 3.26 g - Gypsum (CaSO₄)
^ Brouwmaatje (NL)
10 min - Boil - 1 g - Irish Moss
^ Lot # 4348
^ Brouwstore (NL) 125.249.1
Bottling - 7 items - BrewFerm Carbonation Drops
(33 cl)
^ Brouwmaatje (NL) BM-BL.007.009.33
Bottling - 7 items - 33 cl bottle (26 mm cap)
Long Neck
^ Brouwstore (NL) 017.479.7
Bottling - 7 items - Bottle Caps 26 mm (Orange)
^ Lot # 694-201130-111430-176081-1/1
^ The Malt Miller (UK) EQU-14-007

Yeast

0.5 pkg - Fermentis Safale American US-05
^ Lot # 81025 1949 020
^ Brouwmaatje (NL) BM-BV.40719

Batch Notes

Brew day:

HBC 431 hops were 15.0 % AA in 2024, and are now 10.0 % AA according to hop freshness calculations. Remember to multiply the hop gifts with 1.5 and change the recipe accordingly --> done.

1200 grams of Chateau Pale Ale malt milled with a 0.89 mm mill gap.

Added 3.6 L Spa Reine natural mineral flat water.

Added 0.53 grams canning salt.

Added 0.66 grams Epsom salt.

Added 3.26 grams Gypsum.

Added 1.48 grams Calcium chloride.

Strike water pH = 5.59 @ 11.3 C.

Strike water 72.2 C.

06-04-2026 12:34 started mash tracker.

06-04-2026 12:39 pH = 5.20@41.6 C.

06-04-2026 12:44 pH = 5.25@40.4 C.

06-04-2026 12:49 pH = 5.27@38.8 C.

06-04-2026 12:54 pH = 5.30@36.6 C.

06-04-2026 12:59 pH = 5.29@37.7 C.

06-04-2026 13:04 pH = 5.28@38.4 C.

06-04-2026 13:09 pH = 5.26@41.3 C.

06-04-2026 13:14 pH = 5.31@38.0 C.

06-04-2026 13:19 pH = 5.30@38.0 C.

06-04-2026 13:24 pH = 5.28@39.1 C.

06-04-2026 13:29 pH = 5.30@33.9 C.

06-04-2026 13:34 pH = 5.24@35.6 C.

Hydrometer check SG = 1.001.

Level = 5.0 cm, collected 2.5 L first runnings.

SG = 1.069@42.8 C --> SG = 1.076@20 C.

First batch sparge 2.7 L @ 72 C.

Second batch sparge 2.7 L @ 71 C.

Level is 16.5 cm --> 8.25 L pre boil volume.

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Batch Notes

SG = 1.027 @ 43.7 C --> SG = 1.035 @ 20C.

06-04-2026 15:39 protein break and rolling boil, add 15 minutes to the boil to achieve SG = 1.037.

06-04-2026 15:54 start of the boil tracker.

06-04-2026 16:24 added 4.63 grams of HBC431 hop pellets.

06-04-2026 16:44 added 7.65 grams of HBC431 hop pellets.

06-04-2026 16:44 added 1.00 grams Irish moss.

06-04-2026 16:54 flame out.

Level = 12.7 cm --> 6.35 L post boil volume (hot).

06-04-2026 17:11 added 14.99 grams of HBC431 hop pellets.

SG = 1.036 @ 60.5 C --> SG = 1.052 @ 20 C.

06-04-2026 18:50 split off 1.1 L for batch #54

06-04-2026 19:00 split off 1.1 L for batch #55

Day 0 in fermentation

06-04-2026 20:28 pitched 5.63 grams US-05 yeast at 19.0 C.

06-04-2026 20:28 19.0 C, dp = 0 mm H2O, SG = 1.052, pH = 5.24.

Day 1 in fermentation

07-04-2026 07:45 18.9 C, dp = 50 mm H2O, 3 bubbles.

07-04-2026 18:45 19.0 C, dp = 50 mm H2O, many bubbles.

07-04-2026 22:00 19.0 C, dp = 50 mm H2O, many bubbles.

Day 2 in fermentation

08-04-2026 07:45 18.9 C, dp = 50 mm H2O, many bubbles.

08-04-2026 18:00 18.8 C, dp = 50 mm H2O, many bubbles.

08-04-2026 21:00 18.8 C, dp = 50 mm H2O, many bubbles.

Day 3 in fermentation

09-04-2026 07:30 18.9 C, dp = 50 mm H2O, many bubbles.

09-04-2026 18:15 19.0 C, dp = 50 mm H2O, many bubbles.

09-04-2026 22:30 18.9 C, dp = 50 mm H2O, many bubbles.

Day 4 in fermentation

10-04-2026 07:45 19.1 C, dp = 50 mm H2O, three bubbles.

10-04-2026 11:15 19.4 C, dp = 50 mm H2O, one bubble.

10-04-2026 17:15 18.8 C, dp = 50 mm H2O, no bubbles.

10-04-2026 20:00 19.2 C, dp = 50 mm H2O, one bubble.

Day 5 in fermentation

11-04-2026 07:45 19.0 C, dp = 50 mm H2O, three bubbles.

11-04-2026 16:00 19.6 C, dp = 50 mm H2O, three bubbles.

11-04-2026 20:00 18.8 C, dp = 48 mm H2O, no bubbles.

Day 6 in fermentation

12-04-2026 08:00 18.8 C, dp = 48 mm H2O, no bubbles.

12-04-2026 11:00 18.7 C, dp = 48 mm H2O, no bubbles.

12-04-2026 19:00 19.6 C, dp = 48 mm H2O, no bubbles.

12-04-2026 23:00 19.4 C, dp = 48 mm H2O, no bubbles.

Day 7 in fermentation

13-04-2026 07:30 20.0 C, dp = 48 mm H2O, no bubbles.

13-04-2026 12:00 19.1 C, dp = 45 mm H2O, no bubbles.

13-04-2026 15:45 19.1 C, dp = 45 mm H2O, no bubbles.

13-04-2026 21:00 19.7 C, dp = 45 mm H2O, no bubbles.

Day 8 in fermentation

14-04-2026 07:45 20.0 C, dp = 48 mm H2O, no bubbles.

14-04-2026 19:15 19.1 C, dp = 48 mm H2O, no bubbles.

14-04-2026 21:30 added 19.00 grams HBC 431 hop pellets.

14-04-2026 23:30 19.1 C, dp = 48 mm H2O, no bubbles.

Day 9 in fermentation

15-04-2026 07:30 19.1 C, dp = 48 mm H2O, no bubbles.

15-04-2026 21:00 19.2 C, dp = 45 mm H2O, no bubbles.

Day 10 in fermentation

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Batch Notes

16-04-2026 07:30 19.2 C, dp = 45 mm H2O, no bubbles.
16-04-2026 17:45 20.1 C, dp = 45 mm H2O, no bubbles.
16-04-2026 22:00 19.6 C, dp = 45 mm H2O, no bubbles.

Day 11 in fermentation

17-04-2026 08:00 19.3 C, dp = 45 mm H2O, no bubbles.
17-04-2026 12:00 19.2 C, dp = 45 mm H2O, no bubbles.
17-04-2026 16:00 19.2 C, dp = 45 mm H2O, no bubbles.
17-04-2026 20:00 19.1 C, dp = 45 mm H2O, no bubbles.

Day 12 in fermentation

18-04-2026 08:00 19.2 C, dp = 45 mm H2O, no bubbles.
18-04-2026 17:00 19.0 C, dp = 45 mm H2O, no bubbles.
18-04-2026 21:00 19.5 C, dp = 45 mm H2O, no bubbles.

Day 13 in fermentation

19-04-2026 08:00 19.1 C, dp = 45 mm H2O, no bubbles.
19-04-2026 12:00 19.7 C, dp = 48 mm H2O, no bubbles.
19-04-2026 16:00 19.3 C, dp = 48 mm H2O, no bubbles.
19-04-2026 20:00 19.2 C, dp = 45 mm H2O, no bubbles.

Day 14 in fermentation

20-04-2026 07:45 19.5 C, dp = 45 mm H2O, no bubbles.
20-04-2026 19:00 19.1 C, dp = 45 mm H2O, no bubbles.
20-04-2026 22:00 19.7 C, dp = 48 mm H2O, no bubbles.

Day 15 in fermentation

21-04-2026 07:30 20.0 C, dp = 45 mm H2O, no bubbles.
21-04-2026 19:49 17.6 C, dp = 0 mm H2O, FG = 1.005, pH = 4.13.

Bottling day

21-04-2026 20:15 Filled 7 Long Neck bottles (33 cl).

Day 0 in conditioning

21-04-2026 22:00 17.6 C.

Day 1 in conditioning

22-04-2026 06:30 19.7 C.
22-04-2026 17:30 19.3 C.
22-04-2026 20:00 20.0 C.

Day 2 in conditioning

23-04-2026 07:30 20.0 C.
23-04-2026 18:00 20.0 C.
23-04-2026 21:00 20.0 C.

Day 3 in conditioning

24-04-2026 07:30 20.0 C.
24-04-2026 12:00 20.0 C.
24-04-2026 17:00 19.4 C.
24-04-2026 20:00 19.9 C.

Day 4 in conditioning

25-04-2026 08:00 19.4 C.
25-04-2026 12:00 19.2 C.
25-04-2026 16:00 20.0 C.
25-04-2026 20:00 19.5 C.

Day 5 in conditioning

26-04-2026 08:30 19.7 C.

Extra Measured Values

Strike Water pH 5.59

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Extra Measured Values	
Sparge Water pH	6
Strike Water Temperature	72.2
Sparge Water Temperature	71.5
Pitch Temperature	19

Batch Log	
6 April 2026 11:29	Status: Brewing
6 April 2026	Brew Date
6 April 2026	Fermentation Start
6 April 2026 20:28	SV = 19.5 C.
6 April 2026 20:28	Status: Fermenting
12 April 2026 11:00	SV = 20.0 C
21 April 2026 20:15	Filled 7 Long Neck bottles (33 cl).
21 April 2026	Bottling Date